Problem Set 4: Clustering (Phân cụm dữ liệu)

Comprehensive solutions, reproducible code implementations, mathematical derivations, step-by-step iterations, and dendrograms for **Problem Set 4: Clustering** (Course: Data Mining - AIT3003).

Reference Textbook: *Data Mining: Concepts and Techniques (3rd Edition)* by Jiawei Han, Micheline Kamber, and Jian Pei (Morgan Kaufmann, 2011, Chapter 10: Cluster Analysis: Basic Concepts and Methods).

Table of Contents

1. Problem P1: Cell-Based Entropy for Grid-Based Data
2. Grid Analysis & Point Distribution
3. Shannon Entropy Mathematical Derivation
4. Normalized Entropy and Data Mining Significance
5. Problem P2: K-Means Clustering on 2D Spatial Points
6. Problem Setup & Initial Centroids
7. Iteration 1: Distance Computations, Assignments & New Centroids
8. Iteration 2: Re-assignment & Centroid Updates
9. Iteration 3 & Convergence (Iteration 4)
10. Summary of Answers (Part a & b)
11. Problem P3: Agglomerative Hierarchical Clustering
12. Given Pairwise Distance Matrix
13. Part (a): Single Link (Minimum Distance) Dendrogram
14. Part (b): Complete Link (Maximum Distance) Dendrogram
15. Comparison between Single Link and Complete Link

Problem P1: Cell-Based Entropy for Grid-Based Data

> **Prompt:** Cho một tập dữ liệu gồm 8 điểm dữ liệu phân bố trong lưới 4×4 như hình vẽ. Hãy tính entropy H của phân bố các điểm dữ liệu đó theo phương pháp cell-based entropy.

1. Grid Analysis & Point Distribution

The domain is discretized into an equal-sized 4×4 grid, containing $m = 4 \times 4 = 16$ spatial cells (bins).
Total number of points: $N = 8$.

Counting the points located in each cell (r, c) where $r \in \{1, 2, 3, 4\}$ (rows top to bottom) and $c \in \{1, 2, 3, 4\}$ (columns left to right):

Row \ Column	Column 1	Column 2	Column 3	Column 4	Row Total
Row 1	0	0	0	2	2
Row 2	0	0	0	0	0

Row Column	Column 1	Column 2	Column 3	Column 4	Row Total
Row 3	4	0	0	0	4
Row 4	0	0	2	0	2
Column Total	4	0	2	2	$N = 8$

- There are **3 non-empty cells**:
- $c_1 = \text{Cell}(1, 4)$ contains $n_1 = 2$ points.
- $c_2 = \text{Cell}(3, 1)$ contains $n_2 = 4$ points.
- $c_3 = \text{Cell}(4, 3)$ contains $n_3 = 2$ points.
- The remaining 13 cells contain 0 points.

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2. Shannon Entropy Mathematical Derivation

In grid-based subspace clustering (such as the **ENCLUS** algorithm - *Entropy-based CLUStEring* by Cheng et al., 1999; and Han & Kamber, Chapter 10), the probability (or density ratio) of cell c_i is given by:

$$p_i = \frac{n_i}{N}$$

For empty cells ($n_i = 0$), $p_i = 0$. By standard mathematical convention in information theory:

$$\lim_{p \to 0^+} p \log_2 p = 0$$

For the 3 non-empty cells: - $p_1 = \frac{2}{8} = \frac{1}{4} = 0.25$ - $p_2 = \frac{4}{8} = \frac{1}{2} = 0.50$ - $p_3 = \frac{2}{8} = \frac{1}{4} = 0.25$

Check: $\sum_{i=1}^{16} p_i = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1.0$.

Cell-Based Entropy H (Base 2):

$$H = -\sum_{i=1}^{m} p_i \log_2(p_i) = -\left[p_1 \log_2(p_1) + p_2 \log_2(p_2) + p_3 \log_2(p_3)\right]$$

Substitute the probabilities: - $p_1 \log_2(p_1) = \frac{1}{4}\log_2\left(\frac{1}{4}\right) = \frac{1}{4}(-2) = -0.50$ -
$p_2 \log_2(p_2) = \frac{1}{2}\log_2\left(\frac{1}{2}\right) = \frac{1}{2}(-1) = -0.50$ - $p_3 \log_2(p_3) = \frac{1}{4}\log_2\left(\frac{1}{4}\right) = \frac{1}{4}(-2) = -0.50$

$$H = -\left[-0.50 - 0.50 - 0.50\right] = -(-1.50) = \mathbf{1.5000 \text{ bits}}$$

(If using the natural logarithm $\ln$ base e:
$H_e = -\sum p_i \ln p_i = 2 \times (0.25 \ln 4) + 0.5 \ln 2 = 1.5 \times \ln 2 \approx \mathbf{1.0397 \text{ nats}}$*.)*

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3. Normalized Entropy and Significance

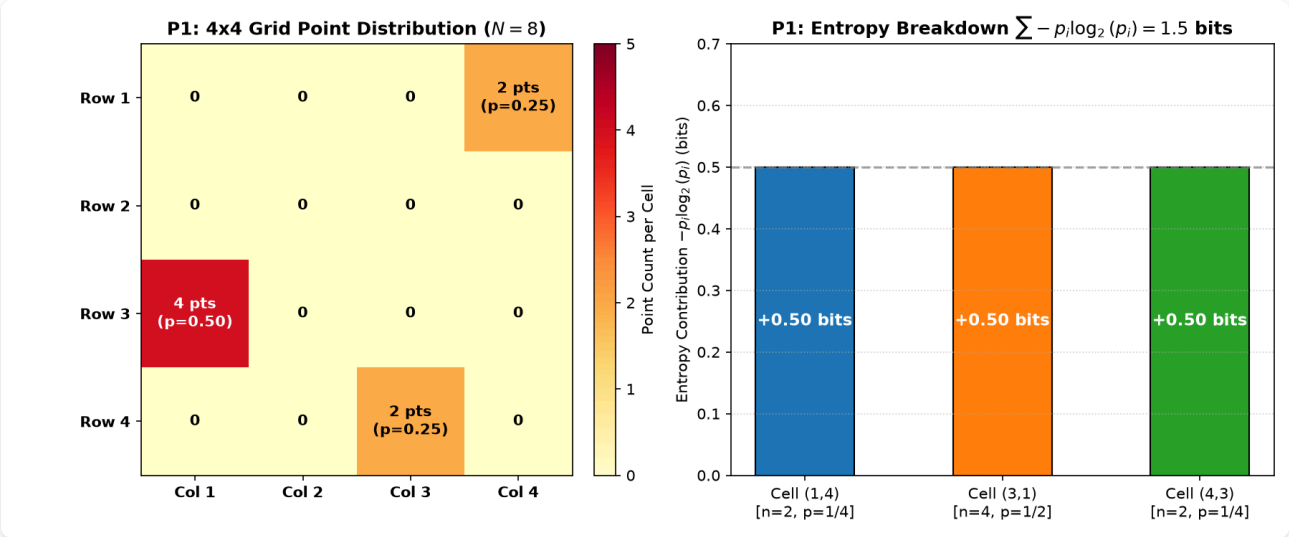
The maximum possible entropy for $m = 16$ cells occurs when points are uniformly spread across all 16 cells ($p_i = 1/16$ for all i):

$$H_{\max} = \log_2(16) = 4.0 \text{ bits}$$

The **normalized cell-based entropy** is:

$$H_{\text{norm}} = \frac{H}{H_{\max}} = \frac{1.50}{4.00} = \mathbf{0.3750} \quad (37.5\%)$$

A low entropy ($\mathbf{1.5 \ll 4.0}$) indicates strong clustering tendency in the subspace: the points are highly concentrated in very few dense grid cells rather than scattered uniformly across the space.



Problem P2: K-Means Clustering on 2D Spatial Points

> **Prompt:** Giả sử rằng nhiệm vụ khai thác dữ liệu là nhóm điểm (với (x, y) biểu diễn vị trí) thành ba cụm, trong đó các điểm là:
> $A_1(2, 10), A_2(2, 5), A_3(8, 4), B_1(5, 8), B_2(7, 5), B_3(6, 4), C_1(1, 2), C_2(4, 9)$.
> Hàm khoảng cách là khoảng cách Euclidean. Giả sử ban đầu chúng ta gán A_1, B_1 và C_1 làm tâm của mỗi cụm, tương ứng. Sử dụng thuật toán k-means để chỉ ra:
> a) Ba tâm cụm sau vòng thực thi đầu tiên.
> b) Ba cụm cuối cùng.

1. Problem Setup & Initial Centroids

- $k = 3$ clusters (C_1, C_2, C_3).
- 8 points:
- $A_1 = (2, 10)$
- $A_2 = (2, 5)$
- $A_3 = (8, 4)$
- $B_1 = (5, 8)$
- $B_2 = (7, 5)$
- $B_3 = (6, 4)$
- $C_1 = (1, 2)$
- $C_2 = (4, 9)$
- Distance function: Euclidean distance $d(\mathbf{p}, \mathbf{m}) = \sqrt{(p_x - m_x)^2 + (p_y - m_y)^2}$.
- Initial seeds:
- $\mathbf{m}_1^{(0)} = A_1 = (2, 10)$
- $\mathbf{m}_2^{(0)} = B_1 = (5, 8)$
- $\mathbf{m}_3^{(0)} = C_1 = (1, 2)$

2. Iteration 1: Distance Computations, Assignments & New Centroids

Compute Euclidean distance from every point $\mathbf{p}$ to $\mathbf{m}_1(2,10)$, $\mathbf{m}_2(5,8)$, and $\mathbf{m}_3(1,2)$:

Point	(x, y)	$d(\mathbf{p}, \mathbf{m}_1)$	$d(\mathbf{p}, \mathbf{m}_2)$	$d(\mathbf{p}, \mathbf{m}_3)$	Minimum Distance	Assigned Cluster
A_1	$(2, 10)$	$\sqrt{0^2 + 0^2} = \mathbf{0.0000}$	$\sqrt{3^2 + 2^2} \approx 3.6056$	$\sqrt{1^2 + 8^2} \approx 8.0623$	0.0000	**Cluster 1**
A_2	$(2, 5)$	$\sqrt{0^2 + 5^2} = 5.0000$	$\sqrt{3^2 + 3^2} \approx 4.2426$	$\sqrt{1^2 + 3^2} \approx \mathbf{3.1623}$	3.1623	**Cluster 3**
A_3	$(8, 4)$	$\sqrt{6^2 + 6^2} \approx 8.4853$	$\sqrt{3^2 + 4^2} = \mathbf{5.0000}$	$\sqrt{7^2 + 2^2} \approx 7.2801$	5.0000	**Cluster 2**
B_1	$(5, 8)$	$\sqrt{3^2 + 2^2} \approx 3.6056$	$\sqrt{0^2 + 0^2} = \mathbf{0.0000}$	$\sqrt{4^2 + 6^2} \approx 7.2111$	0.0000	**Cluster 2**
B_2	$(7, 5)$	$\sqrt{5^2 + 5^2} \approx 7.0711$	$\sqrt{2^2 + 3^2} \approx \mathbf{3.6056}$	$\sqrt{6^2 + 3^2} \approx 6.7082$	3.6056	**Cluster 2**
B_3	$(6, 4)$	$\sqrt{4^2 + 6^2} \approx 7.2111$	$\sqrt{1^2 + 4^2} \approx \mathbf{4.1231}$	$\sqrt{5^2 + 2^2} \approx 5.3852$	4.1231	**Cluster 2**
C_1	$(1, 2)$	$\sqrt{1^2 + 8^2} \approx 8.0623$	$\sqrt{4^2 + 6^2} \approx 7.2111$	$\sqrt{0^2 + 0^2} = \mathbf{0.0000}$	0.0000	**Cluster 3**
C_2	$(4, 9)$	$\sqrt{2^2 + 1^2} \approx 2.2361$	$\sqrt{1^2 + 1^2} \approx \mathbf{1.4142}$	$\sqrt{3^2 + 7^2} \approx 7.6158$	1.4142	**Cluster 2**

Cluster Assignments after Iteration 1:

- $\text{Cluster } 1 = \{A_1\}$
- $\text{Cluster } 2 = \{A_3, B_1, B_2, B_3, C_2\}$
- $\text{Cluster } 3 = \{A_2, C_1\}$

Update Centroids $\mathbf{m}^{(1)}$:

- $\mathbf{m}_1^{(1)} = \frac{(2,10)}{1} = \mathbf{(2, 10)}$
- $\mathbf{m}_2^{(1)} = \frac{(8,4)+(5,8)+(7,5)+(6,4)+(4,9)}{5} = \frac{(30,30)}{5} = \mathbf{(6, 6)}$
- $\mathbf{m}_3^{(1)} = \frac{(2,5)+(1,2)}{2} = \frac{(3,7)}{2} = \mathbf{(1.5, 3.5)}$

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3. Iteration 2: Re-assignment & Centroid Updates

With updated centroids $\mathbf{m}_1^{(1)} = (2, 10)$, $\mathbf{m}_2^{(1)} = (6, 6)$, $\mathbf{m}_3^{(1)} = (1.5, 3.5)$:

Point	(x, y)	$d(\mathbf{p}, \mathbf{m}_1)$	$d(\mathbf{p}, \mathbf{m}_2)$	$d(\mathbf{p}, \mathbf{m}_3)$	Minimum	New Cluster	Shift Note
A_1	$(2, 10)$	$\mathbf{0.0000}$	5.6569	6.5192	0.0000	Cluster 1	Stays

Point					Minimum	New Cluster	Shift Note
A_2	$(2,5)$	5.0000	4.1231	$\mathbf{1.5811}$	1.5811	Cluster 3	Stays
A_3	$(8,4)$	8.4853	$\mathbf{2.8284}$	6.5192	2.8284	Cluster 2	Stays
B_1	$(5,8)$	3.6056	$\mathbf{2.2361}$	5.7009	2.2361	Cluster 2	Stays
B_2	$(7,5)$	7.0711	$\mathbf{1.4142}$	5.7009	1.4142	Cluster 2	Stays
B_3	$(6,4)$	7.2111	$\mathbf{2.0000}$	4.5277	2.0000	Cluster 2	Stays
C_1	$(1,2)$	8.0623	6.4031	$\mathbf{1.5811}$	1.5811	Cluster 3	Stays
C_2	$(4,9)$	$\mathbf{2.2361}$	3.6056	6.0415	2.2361	**Cluster 1**	**Moves from C2 to C1!**

Cluster Assignments after Iteration 2:

- $\text{Cluster } 1 = \{A_1, C_2\}$
- $\text{Cluster } 2 = \{A_3, B_1, B_2, B_3\}$
- $\text{Cluster } 3 = \{A_2, C_1\}$

Update Centroids $\mathbf{m}^{(2)}$:

- $\mathbf{m}_1^{(2)} = \frac{(2,10)+(4,9)}{2} = \mathbf{(3.0, 9.5)}$
- $\mathbf{m}_2^{(2)} = \frac{(8,4)+(5,8)+(7,5)+(6,4)}{4} = \frac{(26,21)}{4} = \mathbf{(6.5, 5.25)}$
- $\mathbf{m}_3^{(2)} = \mathbf{(1.5, 3.5)}$ (unchanged)

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4. Iteration 3 & Convergence (Iteration 4)

With centroids $\mathbf{m}_1^{(2)} = (3, 9.5)$, $\mathbf{m}_2^{(2)} = (6.5, 5.25)$, $\mathbf{m}_3^{(2)} = (1.5, 3.5)$:

- Evaluating $B_1(5, 8)$:
- $d(B_1, \mathbf{m}_1) = \sqrt{(5-3)^2 + (8-9.5)^2} = \sqrt{4 + 2.25} = \sqrt{6.25} = \mathbf{2.5000}$
- $d(B_1, \mathbf{m}_2) = \sqrt{(5-6.5)^2 + (8-5.25)^2} = \sqrt{2.25 + 7.5625} = \sqrt{9.8125} \approx 3.1325$
- Since $2.5000 < 3.1325$, B_1 **moves from Cluster 2 to Cluster 1!**

Cluster Assignments after Iteration 3:

- $\text{Cluster } 1 = \{A_1, B_1, C_2\}$
- $\text{Cluster } 2 = \{A_3, B_2, B_3\}$
- $\text{Cluster } 3 = \{A_2, C_1\}$

Update Centroids $\mathbf{m}^{(3)}$:

- $\mathbf{m}_1^{(3)} = \frac{(2,10)+(5,8)+(4,9)}{3} = \frac{(11,27)}{3} = \left(\frac{11}{3}, 9\right) \approx \mathbf{(3.6667, 9.0000)}$
- $\mathbf{m}_2^{(3)} = \frac{(8,4)+(7,5)+(6,4)}{3} = \frac{(21,13)}{3} = \left(7, \frac{13}{3}\right) \approx \mathbf{(7.0000, 4.3333)}$
- $\mathbf{m}_3^{(3)} = \frac{(2,5)+(1,2)}{2} = \mathbf{(1.5000, 3.5000)}$

Iteration 4 (Verification of Convergence):

Recomputing all distances to $\mathbf{m}_1^{(3)}, \mathbf{m}_2^{(3)}, \mathbf{m}_3^{(3)}$: - Every point is closest to its currently assigned cluster centroid: - $A_1 \to C_1 (d = 1.94)$ - $A_2 \to C_3 (d = 1.58)$ - $A_3 \to C_2 (d = 1.05)$ - $B_1 \to C_1 (d = 1.67)$ - $B_2 \to C_2 (d = 0.67)$ - $B_3 \to C_2 (d = 1.05)$ - $C_1 \to C_3 (d = 1.58)$ - $C_2 \to C_1 (d = 0.33)$ - **No point shifts clusters! The algorithm terminates with full convergence.**

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5. Summary of Answers (Part a & b)

(a) Ba tâm cụm sau vòng thực thi đầu tiên:

$$\mathbf{m}_1^{(1)} = (2, 10)$$

$$\mathbf{m}_2^{(1)} = (6, 6)$$

$$\mathbf{m}_3^{(1)} = (1.5, 3.5)$$

(b) Ba cụm cuối cùng:

- **Cụm 1:** $\{A_1, B_1, C_2\} = \{(2, 10), (5, 8), (4, 9)\}$, tâm cụm $\mathbf{m}_1^* = \left(\frac{11}{3}, 9\right) \approx (3.67, 9.00)$
- **Cụm 2:** $\{A_3, B_2, B_3\} = \{(8, 4), (7, 5), (6, 4)\}$, tâm cụm $\mathbf{m}_2^* = \left(7, \frac{13}{3}\right) \approx (7.00, 4.33)$
- **Cụm 3:** $\{A_2, C_1\} = \{(2, 5), (1, 2)\}$, tâm cụm $\mathbf{m}_3^* = (1.5, 3.5)$



Problem P3: Agglomerative Hierarchical Clustering

> **Prompt:** Bảng dưới đây là ma trận khoảng cách của 6 đối tượng:
> a) Chỉ ra kết quả cuối cùng của phân cụm phân cấp với liên kết đơn (single link) bằng cách vẽ sơ đồ phân cấp (dendrogram).
> b) Chỉ ra kết quả cuối cùng của phân cụm phân cấp với liên kết hoàn chỉnh (complete link) bằng cách vẽ sơ đồ phân cấp.

1. Given Pairwise Distance Matrix

$$D = \begin{bmatrix} & A & B & C & D & E & F \\ A & 0 & & & & & \\ B & 0.12 & 0 & & & & \\ C & 0.51 & 0.25 & 0 & & & \\ D & 0.84 & 0.16 & 0.14 & 0 & & \\ E & 0.28 & 0.77 & 0.70 & 0.45 & 0 & \\ F & 0.34 & 0.61 & 0.93 & 0.20 & 0.67 & 0 \end{bmatrix}$$

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2. Part (a): Single Link (Minimum Distance) Clustering

In **Single Link (Nearest Neighbor)** clustering, the distance between two clusters C_i and C_j is the minimum distance between any pair of their constituent points:

$$d_{\text{single}}(C_i, C_j) = \min_{\mathbf{x} \in C_i,\, \mathbf{y} \in C_j} d(\mathbf{x}, \mathbf{y})$$

Step 1:

- The global minimum entry is $d(A, B) = \mathbf{0.12}$.
- **Merge $\{A\}$ and $\{B\}$** into cluster (A, B) at height $h = 0.12$.
- Update distances:
- $d((A, B), C) = \min(0.51, 0.25) = 0.25$
- $d((A, B), D) = \min(0.84, 0.16) = 0.16$
- $d((A, B), E) = \min(0.28, 0.77) = 0.28$
- $d((A, B), F) = \min(0.34, 0.61) = 0.34$

Step 2:

- Among remaining pairs, the minimum distance is $d(C, D) = \mathbf{0.14}$.
- **Merge $\{C\}$ and $\{D\}$** into cluster (C, D) at height $h = 0.14$.
- Update distances:
- $d((C, D), (A, B)) = \min(d(C, (A, B)), d(D, (A, B))) = \min(0.25, 0.16) = 0.16$
- $d((C, D), E) = \min(0.70, 0.45) = 0.45$
- $d((C, D), F) = \min(0.93, 0.20) = 0.20$

Step 3:

- The minimum distance is $d((A, B), (C, D)) = \mathbf{0.16}$ (via edge $B - D$).
- **Merge (A, B) and (C, D)** into cluster (A, B, C, D) at height $h = 0.16$.
- Update distances:
- $d((A, B, C, D), E) = \min(0.28, 0.45) = 0.28$
- $d((A, B, C, D), F) = \min(0.34, 0.20) = 0.20$
- $d(E, F) = 0.67$

Step 4:

- The minimum distance is $d((A, B, C, D), F) = \mathbf{0.20}$ (via edge $D - F$).
- **Merge (A, B, C, D) and $\{F\}$** into cluster (A, B, C, D, F) at height $h = 0.20$.
- Update distance:
- $d((A, B, C, D, F), E) = \min(0.28, 0.67) = 0.28$ (via edge $A - E$)

Step 5:

- The only remaining pair is (A, B, C, D, F) and $\{E\}$ with distance $\mathbf{0.28}$.
- **Merge into root cluster** (A, B, C, D, E, F) at height $h = 0.28$.

Single Link Merge Sequence Summary:

Step	Clusters Merged	Linkage Distance	New Cluster Formed
1	A and B	$\mathbf{0.12}$	$\{A, B\}$
2	C and D	$\mathbf{0.14}$	$\{C, D\}$
3	$\{A, B\}$ and $\{C, D\}$	$\mathbf{0.16}$	$\{A, B, C, D\}$
4	$\{A, B, C, D\}$ and F	$\mathbf{0.20}$	$\{A, B, C, D, F\}$
5	$\{A, B, C, D, F\}$ and E	$\mathbf{0.28}$	$\{A, B, C, D, E, F\}$

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3. Part (b): Complete Link (Maximum Distance) Clustering

In **Complete Link (Furthest Neighbor)** clustering, the distance between two clusters is the maximum distance between any pair of their constituent points:

$$d_{\text{complete}}(C_i, C_j) = \max_{\mathbf{x} \in C_i,\, \mathbf{y} \in C_j} d(\mathbf{x}, \mathbf{y})$$

Step 1:

- Minimum initial entry is $d(A, B) = \mathbf{0.12}$.
- **Merge** $\{A\}$ **and** $\{B\}$ into cluster (A, B) at height $h = 0.12$.
- Update complete distances:
- $d((A, B), C) = \max(0.51, 0.25) = 0.51$
- $d((A, B), D) = \max(0.84, 0.16) = 0.84$
- $d((A, B), E) = \max(0.28, 0.77) = 0.77$
- $d((A, B), F) = \max(0.34, 0.61) = 0.61$

Step 2:

- Among remaining pairs, the minimum distance is $d(C, D) = \mathbf{0.14}$.
- **Merge** $\{C\}$ **and** $\{D\}$ into cluster (C, D) at height $h = 0.14$.
- Update complete distances:
- $d((C, D), (A, B)) = \max(d(C, (A, B)), d(D, (A, B))) = \max(0.51, 0.84) = 0.84$
- $d((C, D), E) = \max(0.70, 0.45) = 0.70$
- $d((C, D), F) = \max(0.93, 0.20) = 0.93$

Step 3:

Current pairwise distances between clusters (A, B), (C, D), E, and F: - $d((A, B), F) = \mathbf{0.61}$ - $d(E, F) = 0.67$ - $d((C, D), E) = 0.70$ - $d((A, B), E) = 0.77$ - $d((A, B), (C, D)) = 0.84$ - $d((C, D), F) = 0.93$

Minimum distance is $d((A,B),F) = \mathbf{0.61}$. - **Merge (A,B) and $\{F\}$** into cluster (A,B,F) at height $h = 0.61$. - Update distances: - $d((A,B,F),E) = \max(d((A,B),E), d(F,E)) = \max(0.77, 0.67) = 0.77$ - $d((A,B,F),(C,D)) = \max(d((A,B),(C,D)), d(F,(C,D))) = \max(0.84, 0.93) = 0.93$

Step 4:

Current distances between clusters (A,B,F), (C,D), and E: - $d((C,D),E) = \mathbf{0.70}$ - $d((A,B,F),E) = 0.77$ - $d((A,B,F),(C,D)) = 0.93$

Minimum distance is $d((C,D),E) = \mathbf{0.70}$. - **Merge (C,D) and $\{E\}$** into cluster (C,D,E) at height $h = 0.70$. - Update distance: - $d((A,B,F),(C,D,E)) = \max(d((A,B,F),(C,D)), d((A,B,F),E)) = \max(0.93, 0.77) = \mathbf{0.93}$

Step 5:

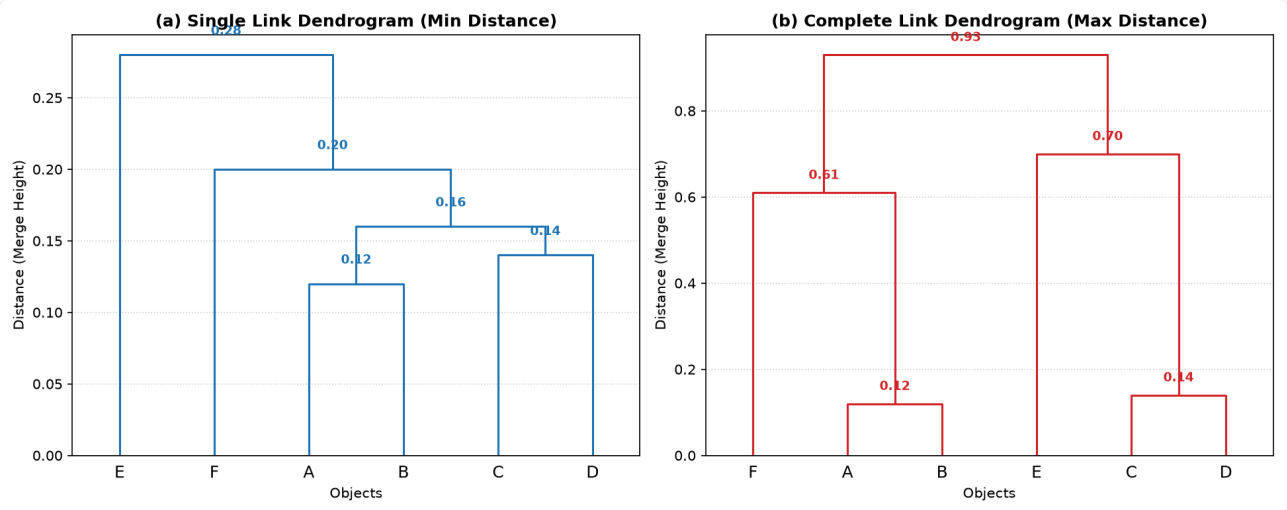
- **Merge (A,B,F) and (C,D,E)** into root cluster (A,B,C,D,E,F) at height $h = \mathbf{0.93}$.

Complete Link Merge Sequence Summary:

Step	Clusters Merged	Linkage Distance	New Cluster Formed
1	A and B	$\mathbf{0.12}$	$\{A,B\}$
2	C and D	$\mathbf{0.14}$	$\{C,D\}$
3	$\{A,B\}$ and F	$\mathbf{0.61}$	$\{A,B,F\}$
4	$\{C,D\}$ and E	$\mathbf{0.70}$	$\{C,D,E\}$
5	$\{A,B,F\}$ and $\{C,D,E\}$	$\mathbf{0.93}$	$\{A,B,C,D,E,F\}$

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4. Comparison between Single Link and Complete Link



1. **Chaining Effect in Single Link:**
2. Single link creates elongated, chained structures ($((((A+B)+(C+D))+F)+E)$) because two clusters merge as soon as a single close point pair is found ($d(B,D) = 0.16$, $d(D,F) = 0.20$, $d(A,E) = 0.28$).
3. The tree height is much more compressed (maximum merge height is only 0.28).

4. **Compact Spherical Clusters in Complete Link:**
5. Complete link requires *all* pairs between two clusters to be close. Because $d(A, D) = 0.84$ and $d(C, F) = 0.93$, $\{A, B\}$ and $\{C, D\}$ cannot merge early.
6. Instead, $\{A, B\}$ absorbs F to form $\{A, B, F\}$, while $\{C, D\}$ absorbs E to form $\{C, D, E\}$, producing two balanced, compact trios before finally uniting at $h = 0.93$.